



THE PREDICTIVE RADAR: INTERCEPTING CUSTOMER CHURN AT SYRIATEL

Deploying machine learning to identify at-risk customers before they
leave

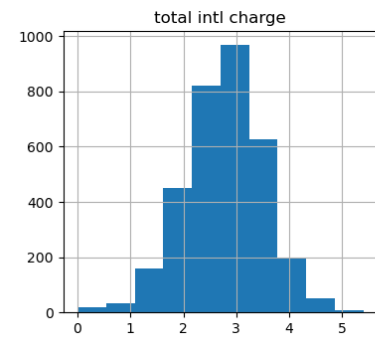
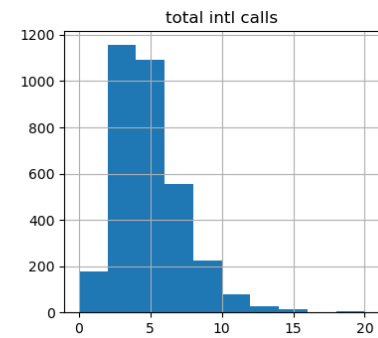
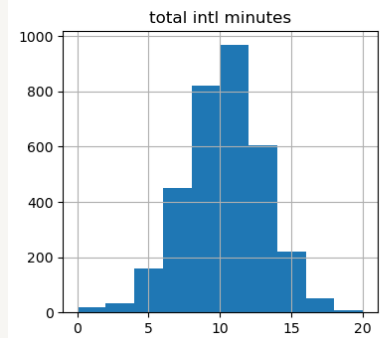
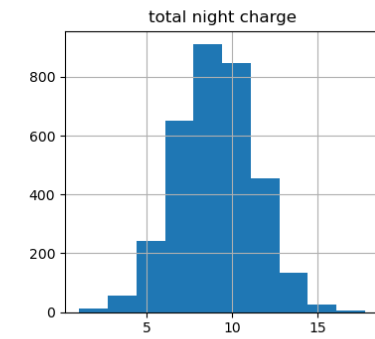
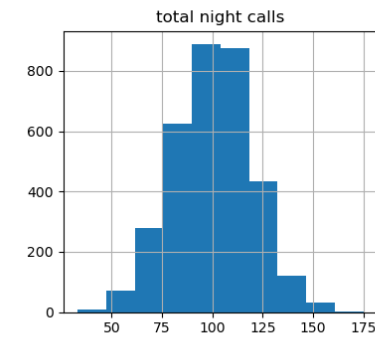
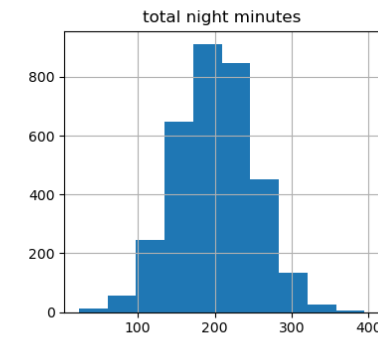
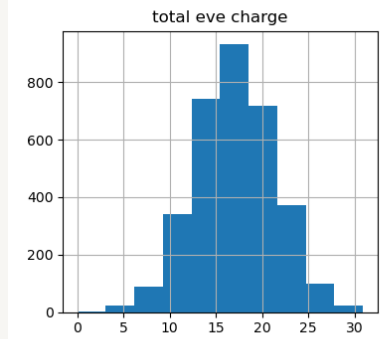
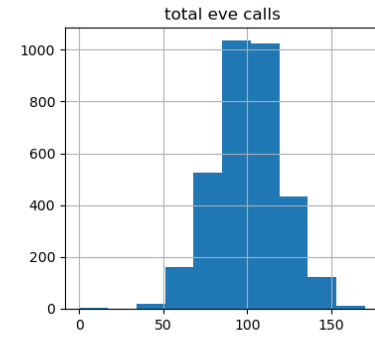
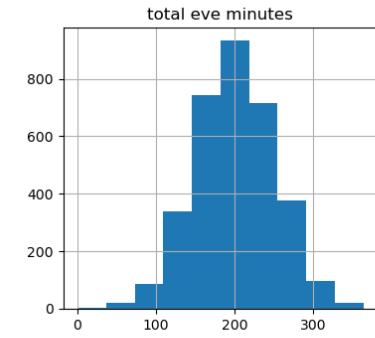
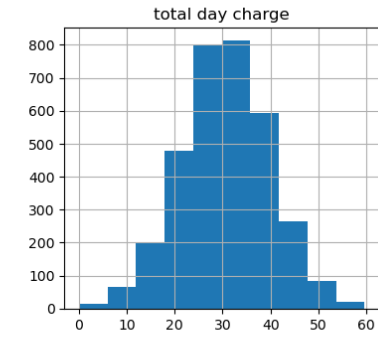
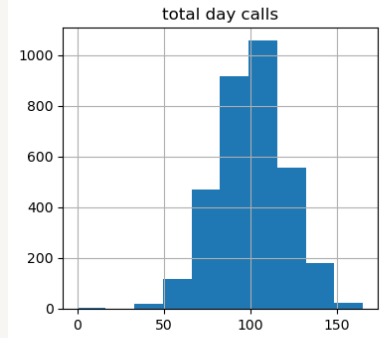
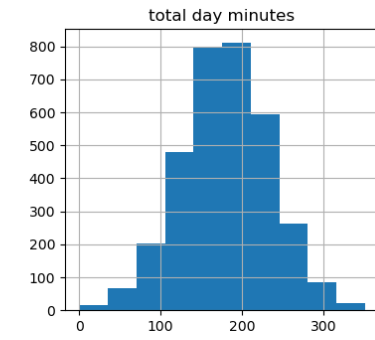
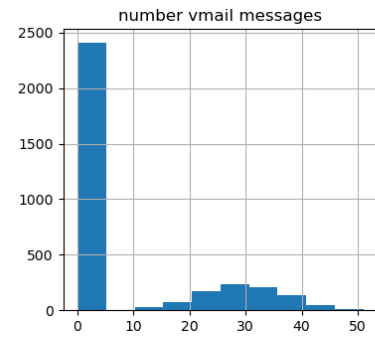
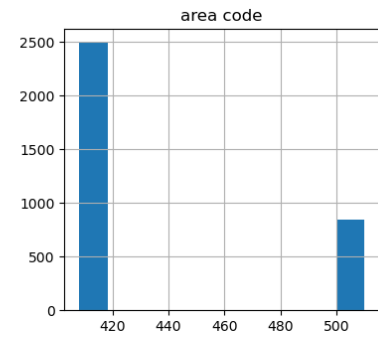
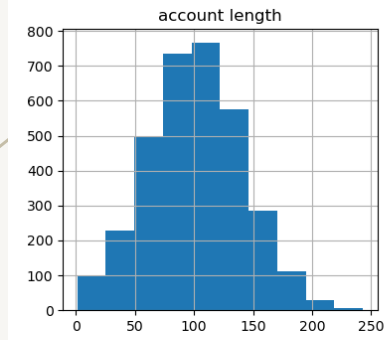
Bennie Kibet

BUSINESS UNDERSTANDING

- Syriatel is successfully growing its customer base, offering diverse packages including highly adopted international and voicemail plans.
- The company has experienced growth but has a challenge in retaining customers, as they opt out of its services (churn).
- Therefore, acquiring new customers is typically more expensive than retaining existing ones, so reducing churn is a high-priority business goal.

DATA UNDERSTANDING

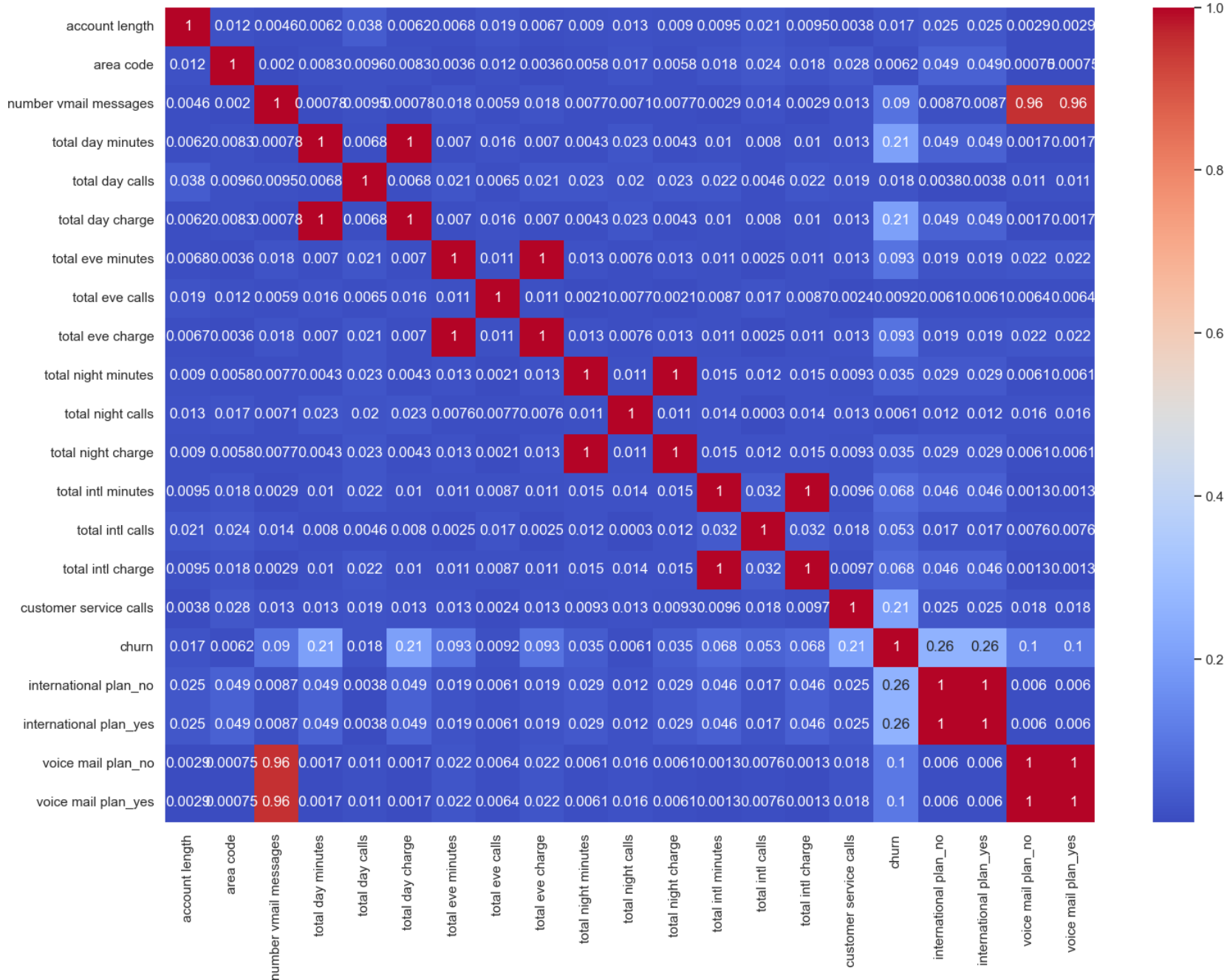
- The data set contained 21 features with 3333 data entries of customers.
- The data had no missing or null values.



DISCUSSION

- The distributions show that most usage-related variables (e.g., total day/evening/night minutes and charges) are approximately bell-shaped, suggesting stable and consistent customer behavior across time periods.
- Variables such as the number of voicemail messages and customer service calls are highly right-skewed, with many customers having low values and a few having very high counts.
- Call counts (day/evening/night) are moderately normally distributed, reflecting consistent calling patterns, while international calls and minutes are skewed toward lower values, indicating fewer users engage heavily in international usage.

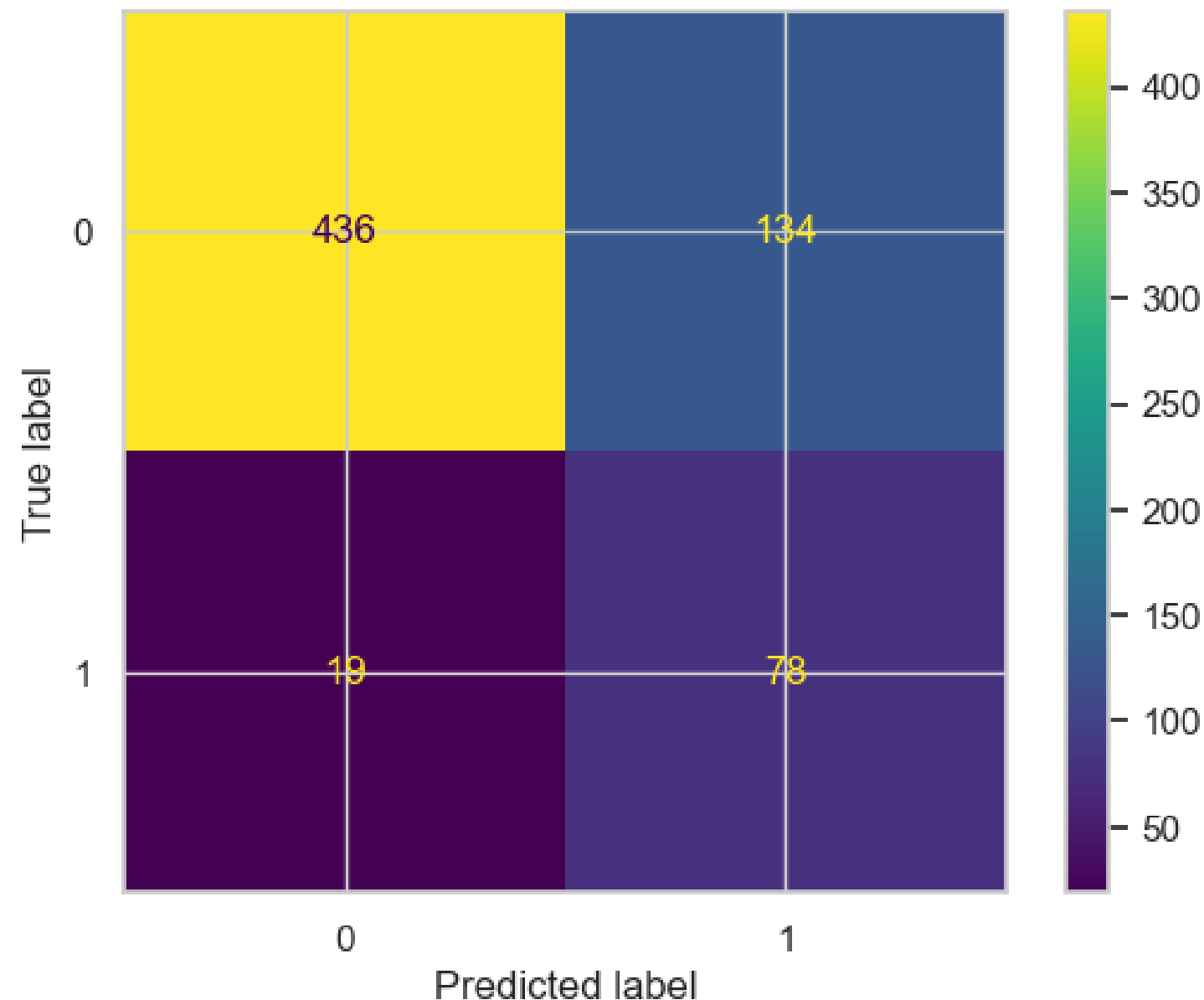
Correlation Matrix of Syriatel Churn



DISCUSSION

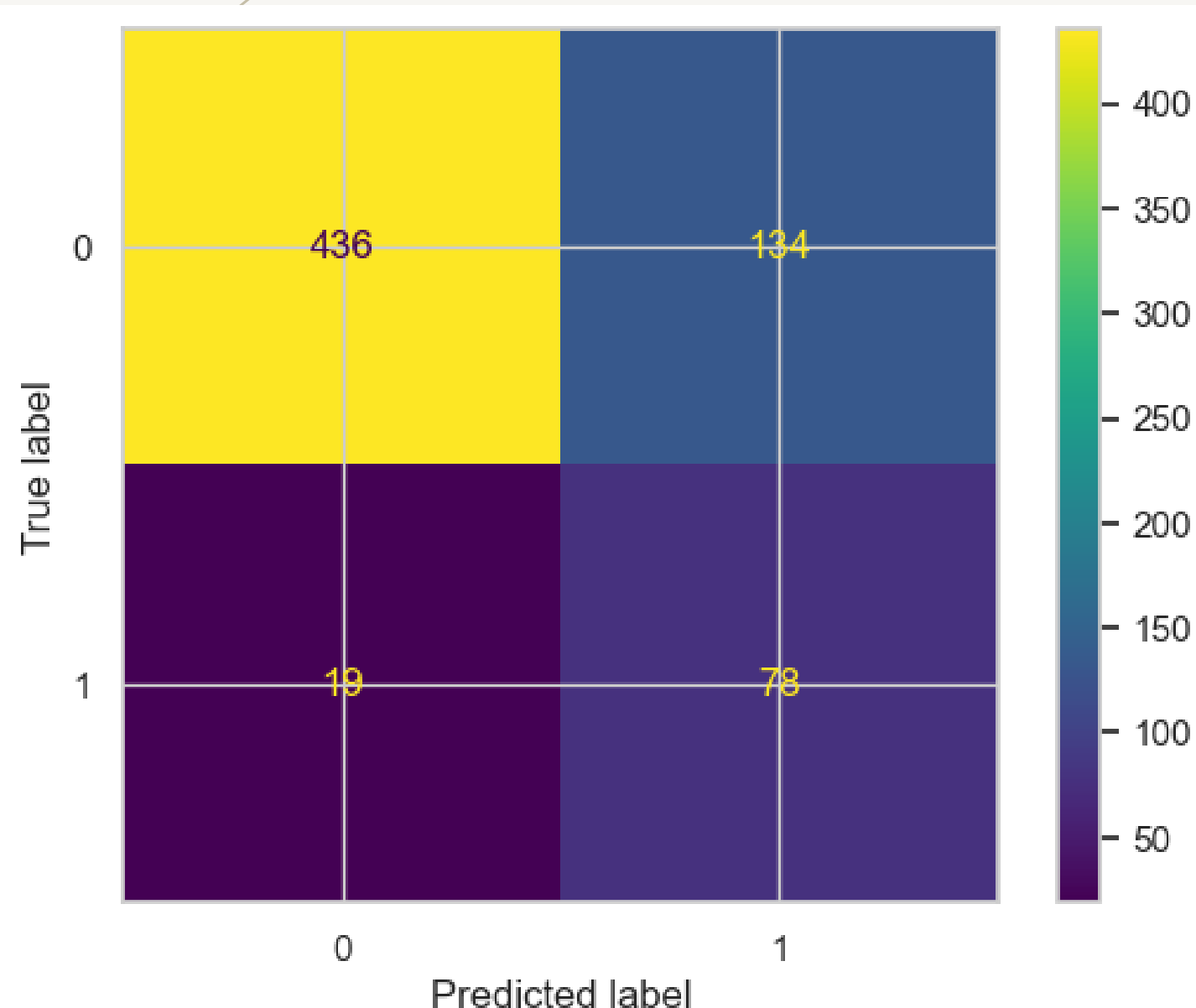
- The correlation matrix shows strong multicollinearity between usage variables and their corresponding charges (e.g., total minutes and total charges are almost perfectly correlated), indicating redundancy.
- Most features have weak correlation with churn, suggesting no single variable strongly predicts customer loss.
- The most notable relationships with churn are customer service calls, international plan, and total day minutes and day charges, though correlations are still moderate.
- Binary features like international plan and voicemail plan are highly correlated with each other due to encoding.
- Overall, the model will need to rely on combined feature effects rather than individual predictors for accurate churn prediction.

BASELINE MODEL: DISCUSSION



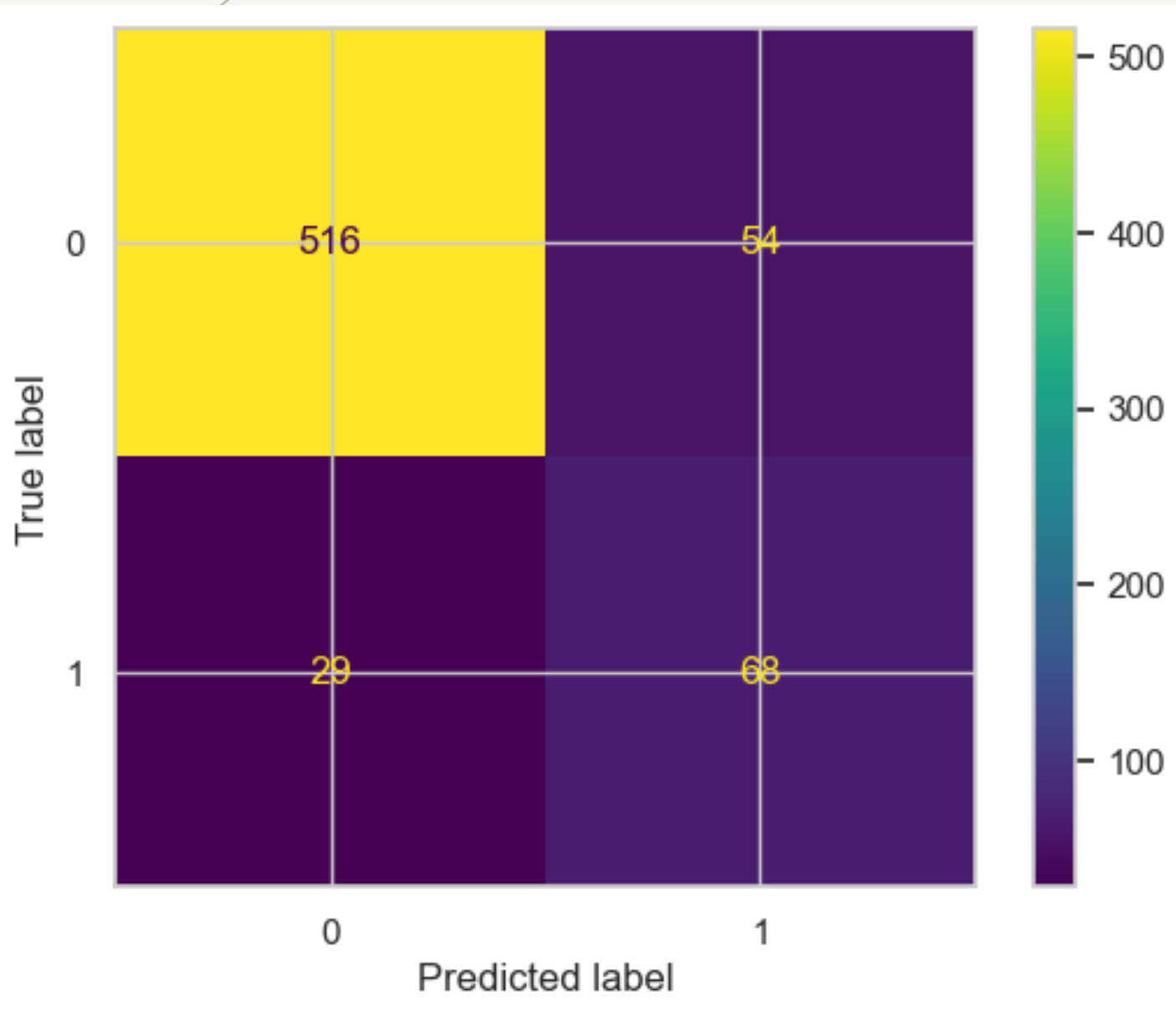
- The model correctly predicts most class 0 instances (440 correct vs 130 misclassified), indicating strong performance on class 0.
- For class 1, performance is weaker, with 76 correct predictions and 21 misclassified as class 0.
- There is a noticeable bias toward predicting class 0, as seen from higher correct and incorrect counts in that column.
- False positives (130) are relatively high compared to false negatives (21), suggesting overprediction of class 1 when the true class is 0.
- Overall, the classifier performs better on class 0 than class 1, indicating class imbalance or model bias.

MODEL 2: DISCUSSION



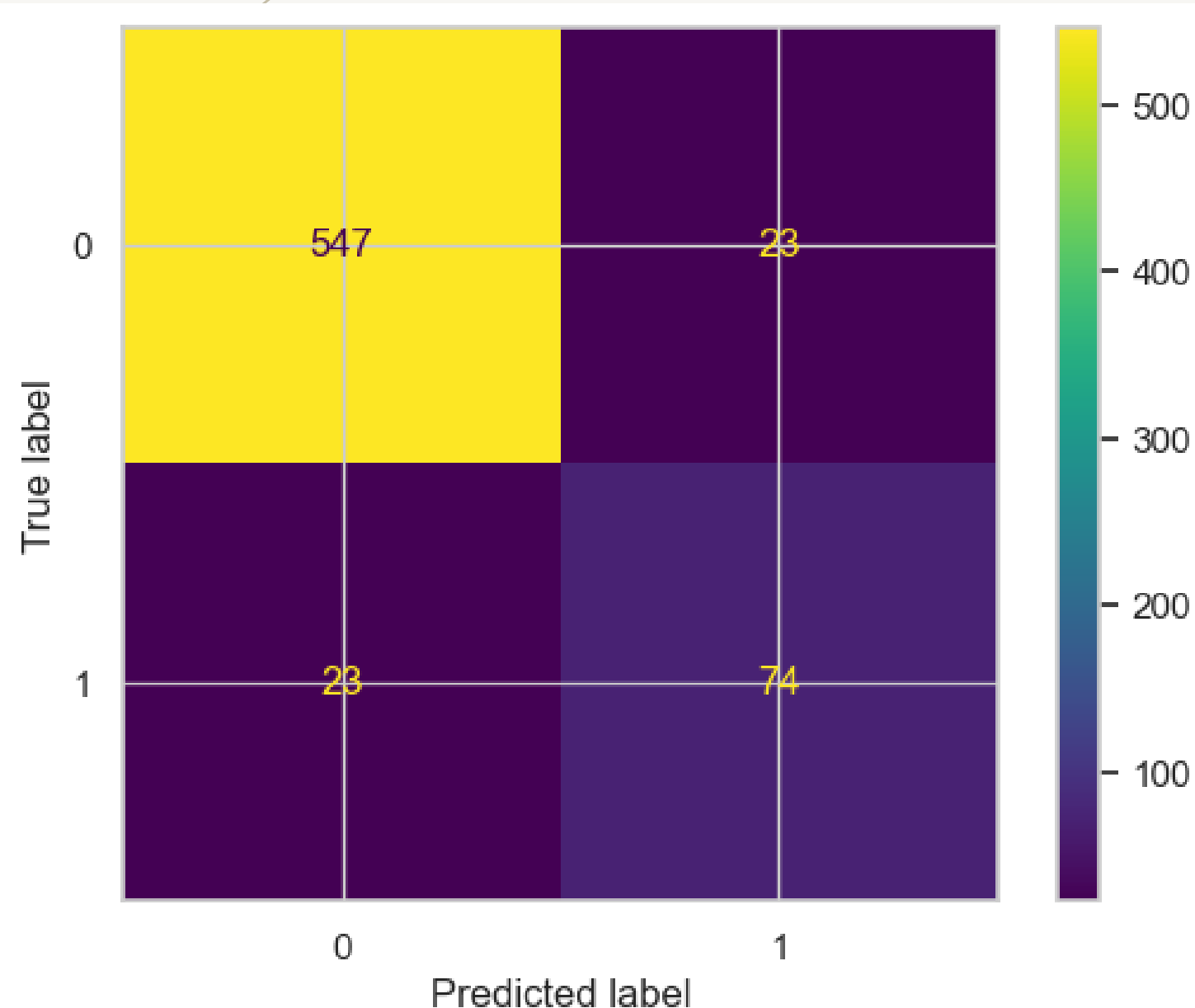
- The model still performs well on class 0 (436 correct vs 134 misclassified), but errors remain relatively high.
- Class 1 detection is modest, with 75 correct predictions and 19 false negatives.
- False positives (134) are significantly higher than false negatives (19), indicating a tendency to misclassify class 0 as class 1.
- Compared to the previous matrix, performance is nearly unchanged, with a slight drop in correct class 0 predictions and a minor increase in misclassification.

MODEL 3: DISCUSSION



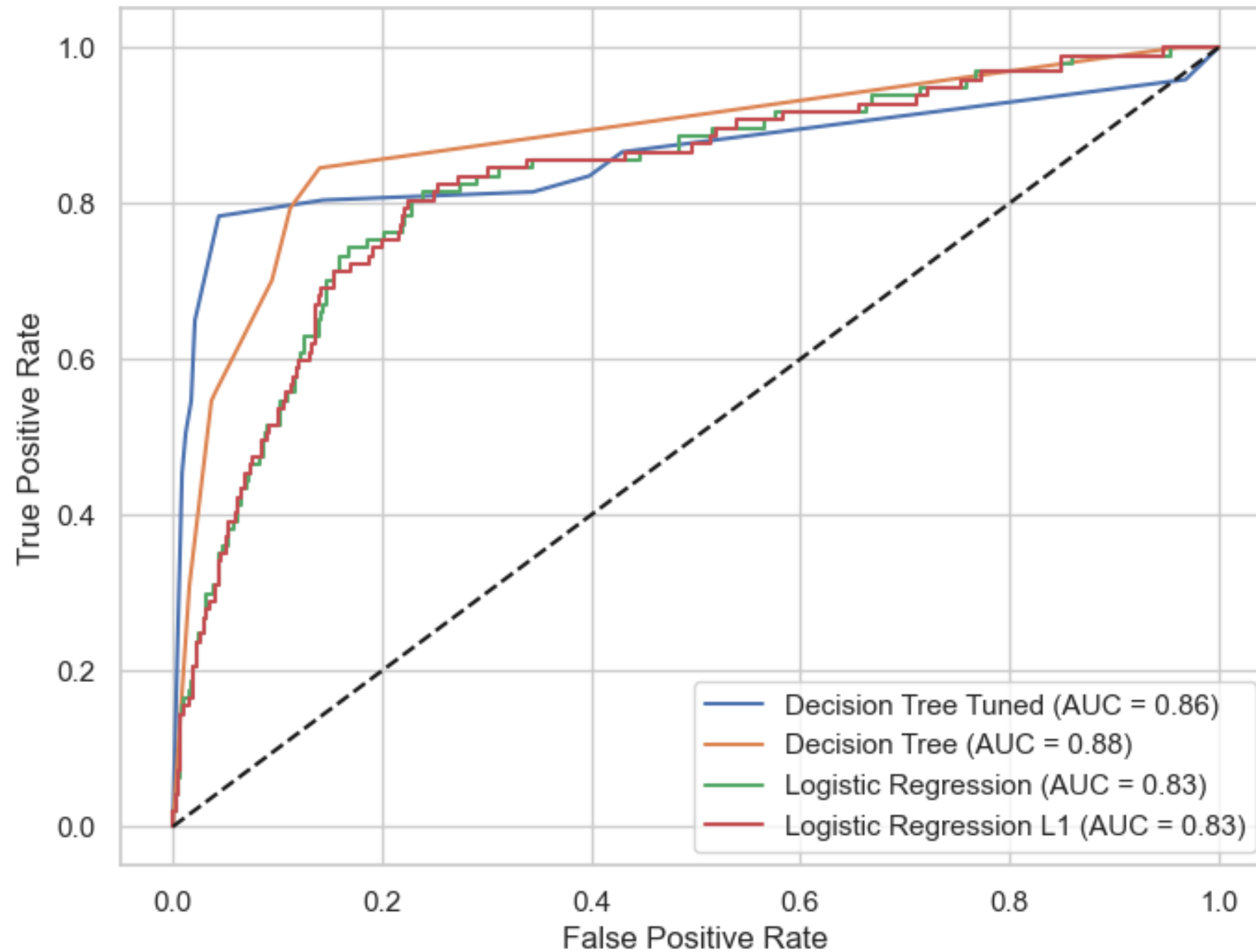
- The model shows strong improvement in class 0 prediction (516 correct vs 54 misclassified), significantly reducing false positives.
- Class 1 performance is stable, with 68 correct predictions and 29 false negatives.
- False positives (54) are now much lower than in previous matrices, indicating better precision for class 1 predictions.

MODEL 4: DISCUSSION



- The model shows strong improvement in class 0 prediction (547 correct vs 23 misclassified), significantly reducing false positives.
- Class 1 performance is stable, with 74 correct predictions and 23 false negatives.
- False positives (23) are now much lower than in previous matrices, indicating better precision for class 1 predictions.

ROC Curve



DISCUSSION

- The baseline decision tree performs best (AUC = 0.88), indicating the strongest class separation.
- The tuned decision tree is slightly worse (AUC = 0.86), suggesting tuning may have reduced overfitting at a small cost to accuracy.
- Both logistic regression models perform similarly (AUC = 0.83) and trail the trees.
- This implies nonlinear relationships in the data favor tree-based models.
- All models outperform random guessing, showing meaningful predictive power.
- Overall, the decision tree model is the most reliable here, while the others may need better tuning, features, or different modeling approaches.

CONCLUSION

- Accurate Prediction of churns: The model's high accuracy ensures effective identification of customers likely to churn, enabling preparation on how to appease the customer.
- By integrating this model into their customer analytics pipeline, Syriatel can shift from reactive to preventive retention strategies, targeting users before they leave.
- The model's solid recall ensures most potential churners are detected, while its overall accuracy supports confident decision-making.
- Although some false positives exist, they are acceptable in a retention context where early engagement is preferable to losing customers.

RECOMMENDATIONS

- Deploy the model in real-time or periodic scoring systems to continuously monitor churn risk.
- Design targeted retention campaigns (discounts, loyalty rewards, personalized offers) for high-risk customers.
- Improve feature engineering by incorporating usage patterns, complaints, and customer service interactions.
- Regularly retrain and validate the model to maintain performance as customer behavior evolves.
- Combine model outputs with business rules (e.g., high-value customers) to prioritize intervention efforts.

A series of thin, light-brown lines forming an abstract geometric pattern in the top-left corner of the slide. The lines intersect to create various triangular and polygonal shapes.

THANK YOU

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